On the Universal Structure of Maximal Planar Graphs

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Abstract

This paper proves that every maximal planar graph admits a composition series analogous to that of a solvable group. The role of normal subgroups is played by a special class of maximal planar graphs. Each graph in this class contains a maximal outerplanar subgraph with the same number of vertices. Furthermore, we prove that every graph in this class is four-colorable.

1 Introduction

This paper begins with an investigation of coloring methods for planar graphs. For a planar graph, if we perform a chain-like coloring from a local region, it is always impossible to exclude the possibility of conflicts arising somewhere in the process. To avoid such conflicts, one must find a "unidirectional" coloring method for planar graphs.

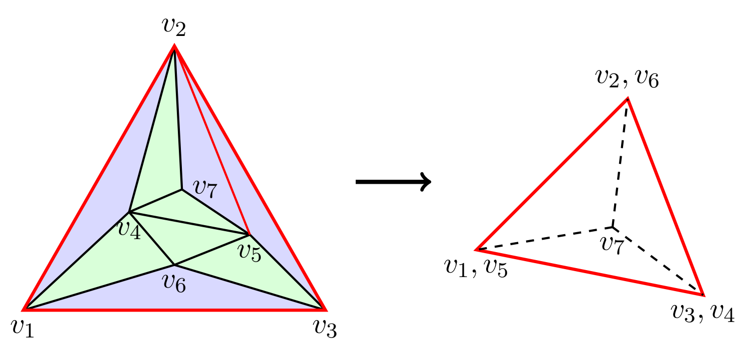
Consider a K_3 in a maximal planar graph. The coloring of the internal vertices of this K_3 is independent of the coloring outside the

K_3. Motivated by this observation, if any maximal planar graph in which every K_3-face contains no interior vertices is four-colorable, then the unidirectional coloring of all maximal planar graphs follows.

Further analysis shows that this condition can be relaxed. If G contains a maximal outerplanar subgraph with the same number of vertices is four-colorable, then such a unidirectional coloring is still holds (we call such graphs *Normal Graphs*).

Based on this idea, we prove that every maximal planar graph is a normal graph, and that there exists a composition series analogous to that of solvable groups.

Finally, we will show that all normal graphs are 4-colorable. As shown in the figures below.



1.1 1.2

Figure 1.1 is divided into two parts, marked by blue and green shading. These two parts each have the same number of vertices 7, edges 11, and faces 6. We denote by $\overline{S_4}$ the graph of figure 1.1. And the green part is denoted by $\overline{R_4}$.

Consider the graph $\overline{R_4}$ and figure 1.2. Clearly, $\overline{R_4}$ can be **folded** into K_4 (Here, folding is understood in the sense of origami). And the number of ways to fold it is 2^4. We will prove that one of these ways of folding defines a homomorphism from $\overline{S_4}$ to K_4.

This process involves only two difficulties. First, whether the red edges in $\overline{S_4}$ may collapse to a single vertex. Second, assuming the first condition is satisfied, whether there exists a vertex that cannot be folded into K_4.

Regarding the existence of the red edges in $\overline{S_4}$ as constraints. If $\overline{R_4}$ cannot be folded into K_4 then each vertex of the K_4 shown in Figure 1.2 corresponds to a nonempty set. Every element of each set is adjacent to at least one element in each of the other three nonempty sets. Thus, we obtain a K_4-minor and hence a K_5-minor which contradicts the fact that $\overline{S_4}$ is a planar graph. A proof of this result will be given in Theorem 4.

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We now generalize the conclusion of the first part to all planar graphs. We begin by recursively defining the maximal outerplanar graphs considered in the first part. To do this, we need to use the concept of "gluing" from topology.

Definition 1. *Let $\{u_i\}$ be a set whose elements are K_3's. G is called a maximal outerplanar graph, denoted by $\overline{R_n}$. Define as follows*

1. *$\overline{R_0}$ is u_0.*
2. *$\overline{R_1}$ is obtained by gluing a K_2 in u_0 and a K_2 in u_1 together from the outside.*
3. *$\overline{R_n}$ is obtained by gluing a K_2 that hasn't been glued yet in $u_{i,\ i<n}$ and a K_2 in u_n together from the outside.*

Definition 2. *Normal Graph*

For any maximal planar graph G, if G contains a subgraph $\overline{R_n}$ with the same number of vertices as G, then G is called a normal graph denoted by $\overline{S_n}$. In particular, if every K_3-face in G has no vertices in its interior, then G is called a minimal normal graph.

From the definitions, $\overline{R_n}$ have $n+3$ vertices, $2n+3$ edges, and $n+2$ faces. We assert that minimal $\overline{S_n}$ is the fundamental structure underlying every maximal planar graph.

Theorem 1. *Every maximal planar graph G is a $\overline{S_n}$.*

Proof. All empty K_3-faces in G form a finite sequence $\{u_k\}$. Starting from u_0, we traverse the sequence and construct $\overline{R_n}$ according to Definition 1.

If the intersection $u_i \cap \overline{R_{i-1}}$ is empty, then we move u_i to the end of the sequence. Whenever $u_i \cap \overline{R_{i-1}}$ is neither empty nor equal to K_2, we simply omit it.

Repeating this procedure, we eventually obtain $\overline{R_n}$ containing all vertices of G. The connectedness of G guarantees that this construction is valid.

□

Theorem 2. *For any maximal planar graph G whose order is bigger than 3, there exists a minimal normal subgraph $\overline{S_n}$ such that $n > 0$ and $\overline{S_n}$ contains the three outermost vertices of G.*

Proof. We denote by $V'(G)$ the set consisting of all vertices of G **except its three outermost vertices**. Since G is a maximal planar graph, for any one of its edges, there exist several vertices in $V'(G)$ such that each of these vertices is adjacent to the two endpoints of that edge, thereby forming a sequence of $K_3's$ that are nested by inclusion.

Keep the largest one of this sequence of $K_3's$ and delete all vertices and edges in its interior. If this vertex is connected to all three outermost vertices, then the resulting subgraph is $\overline{S_1}$.

Apply the above procedure repeatedly until no K_3-face of G contains any interior vertex. The graph obtained in this way is the $\overline{S_n}$ we are looking for.

$\square$

By Theorem 2, every maximal planar graph admits a composition series analogous to that of a solvable group.

Definition 3. *Let G be a maximal planar graph. $\overline{S_n}$ is the minimal normal subgraph of G. Module of G, denoted by $G^{(j)}$ is defined as follows*

1. *$G^{(0)}$ is $\overline{S_n}$.*
2. *$G^{(j)}$ is the union of $G^{(j-1)}$ and all minimal $\overline{S_n}$ corresponding to the subgraphs associated with each empty K_3-face in $G^{(j-1)}$.*

We have

Theorem 3. *G is a maximal planar graph if and only if there exists a composition series as follows.*

$$\overline{S_n} = G^{(0)} \vartriangleright G^{(1)} \vartriangleright G^{(2)} \vartriangleright \cdots \vartriangleright G^{(j)} = G$$

Proof. The sufficiency has already been established in the above definition; we only prove the necessity.

By Definition 3, each newly added part in the composition series is a union of several $\overline{S_n}$, where the outermost K_3 of each $\overline{S_n}$ is exactly a K_3 in some $\overline{S_n}$ from the previous term. Hence, G is planar.

$\square$

We are now ready to prove the main theorem of this paper.

Theorem 4. *For any $\overline{S_n}$, there exists a homomorphism from $\overline{S_n}$ to K_4.*

Proof. Assume that for all 2^n ways of folding $\overline{R_n}$, there always exists a vertex that cannot be folded into K_4 due to the restriction of $\overline{S_n}$. We shall show that this assumption implies that $\overline{S_n}$ has a K_5-minor.

Consider an arbitrary folding $\phi : \overline{R_n} \to K_4$. Suppose that a vertex v cannot be folded into K_4 under ϕ. For every vertex x in K_4, there exists a set X such that X consists of all preimages of x that are adjacent to v. Since each vertex has two choices during the folding process, some elements in X under ϕ can still be moved to other vertices of K_4 without creating conflicts. We denote by X' the set consisting of those vertices that **cannot** be moved under ϕ. By the assumption that none of the 2^n folding schemes defines a homomorphism from $\overline{S_n}$ to K_4. Sets X'_1, X'_2, X'_3 and X'_4 have the following properties:

1. **X'_i is nonempty.**
2. **Every element of X'_i is adjacent to at least one element in each of other three sets.**

Consider any three of the four sets, say X'_1, X'_2, and X'_3. For each element in X'_i, we select one element in X'_j that is adjacent to it. Thereby defining a cyclic contraction mapping.

$$X'_1 \to X''_2 \to X''_3 \to X''_1 \cdots$$

Each term in the sequence is the image of the previous term under the mapping. This sequence converges to a cyclic **bijection** among three sets denote by X^*_1, X^*_2 and X^*_3. The vertices in these three sets form at least one cycle, and hence a K_3-minor.

The fact that any three of the sets X^*_i form a K_3-minor implies the existence of four K_3-minors, corresponding to the four triangular faces of a K_4. Even if the cycles yielding these K_3-minors may be pairwise vertex-disjoint, they still connected in $\overline{S_n}$. That means there is a K_4-minor in $\overline{S_n}$. Since every vertex of this K_4-minor is adjacent to v, we find a K_5-minor. This contradicts the fact that $\overline{S_n}$ is planar.

Thus, among the 2^n ways of folding $\overline{R_n}$, at least one admits a homomorphism from $\overline{S_n}$ to K_4.

$\square$

Combining Theorem 1 and Theorem 4, it follows that every maximal planar graph is 4-colorable.

3

We now state the direct condition underlying the above proof. This condition is not that G has no K_5-minor, but that G is planar.

Definition 4. *as follows*

A graph G is said to admit a crossing-free embedding into $\mathbb{R}^{m-2}$ if there exists a mapping of G into $\mathbb{R}^{m-2}$ such that for any $(m-2)$-polytope realized by a subgraph H, no edge of G crosses it.

Using Definition 4, we can easily define maximal graphs in $\mathbb{R}^{m-2}$.

Definition 5. *as follows*

A graph G is called a maximal graph if adding any edge to G destroys the existence of a crossing-free embedding of G into $\mathbb{R}^{m-2}$.

If the following proposition holds. Then all definitions and theorems in Section 2 extend directly to arbitrary natural numbers $m \geq 4$.

If G admits a crossing-free embedding into $\mathbb{R}^{m-2}$ then G has no K_{m+1}-minor.

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